

Term	譯名	Definition
Revenue, sales, turnover	營業收入	Total amount of income generated by the company's goods or service.
Gross profit, profit, gross income	營業毛利	Revenue minus cost of revenue or direct cost.
Operating income	營業利潤	Gross profit minus operating expense.
Interest income	利息收入	Amount of money obtained from investments that pay interest, such as a savings account or certificate of deposit.
Earnings before income tax (EBIT), income before income tax (IBIT)	稅前淨利	Operating income plus other income and minus non-operating expense.
Net income, net earnings	稅後淨利	EBIT minus tax income expenses or benefit.
Cost of revenue	營業成本	Direct cost related to selling goods or providing service.
Operating expense (OpEx)	營業費用	Indirect cost such as insurance, office supplies, utilities, research, marketing fee, and rent.
Capital expenditure (CapEx)	資本支出	Purchases of significant goods or services
Interest expense	利息支出	Interests of loans, debts, or borrowings.

Table 9: Definitions of terms related to income statements.

Relative valuation refers to popular multiples or ratios that can be easily calculated from a company's financial numbers. This include,

- P/E ratio
- PEG ratio
- P/B ratio
- P/FCFE ratio
- EV/EBTIDA ratio
- EV/CFO ratio
- FCFE/EV ratio
- Debt-to-equity ratio
- Dividend yield
- Free cash flow yield
- Buyback yield
- Return on equity
- Return on invested capital
- Market implies growth appreciation period

Limitations of relative valuation Since relative valuation relies on multiples and ratios that only take a part of a company's financial statements into consideration, often they do not paint the whole picture and do not take any future events into consideration. By comparing multiples of ratios with companies or broad market, we also assume that the companies have the same marketing condition and cost structures. Lastly, relative valuation cannot be used on companies from different industries; different industries have different earning profiles.

Valuation multiples When multiples use the value measure of a company as its numerator, its called a *valuation multiple*. Depending on the type of value measure, multiples are categorized as either equity value multiple or enterprise value multiple. The denominator of multiples are thus appropriate value drivers for each kind of value measure.

$$\text{Valuation multiple} = \frac{\text{Value measure}}{\text{Value driver}}$$

Value measure	Value driver	Valuation multiple
Equity value	Earnings	P/E ratio
	Book value	P/B ratio
Enterprise value	Free cash flow to equity	P/FCFE ratio
	EBITDA	EV/EBITDA ratio
	Revenue	EV/Revenue ratio
	Free cash flow to the firm	EV/FCFF ratio

Table 10: Valuation multiples using equity value and enterprise value as value measures and their respective appropriate value drivers.

4.2.1 Equity Value Multiples

P/E Ratio

Price-to-earnings ratio is the current market price of a company's share divided by its EPS. It is also called the price multiple or earnings multiple. P/E represents a the relative valuation of a stock and is best used when compared with past P/E ratios, with companies in the same industry, or with a broader market such as S&P 500 Index.

Since EPS is readily available in a company's income statement, P/E is the most popular choice of valuation. There are two kinds of P/E ratios depending on what kind of EPS value is being used:

1. Trailing P/E ratio: TTM EPS is used.
2. Forward P/E ratio: EPS value is based on estimated future earnings.

Forward P/E ratio uses estimated future earnings which vary greatly between analysts and are often overoptimistic. In terms of predicting the future performance, trailing P/E ratio is better than forward P/E ratio [9].

A P/E ratio below a certain value or above a certain value does not automatically make it undervalued or overvalued. A high P/E ratio could signal that the investors expect the company to grow, and a low P/E ratio could also mean that the company is facing difficulties. When EPS is close to zero or below zero, P/E ratio could fluctuate violently and become inapplicable.

P/E ratio does not mean the time it takes for investors to get back their revenue, despite it being a popular analogy. While a P/E ratio of 5 and share price of \$20 means that earnings are \$4 per year and it takes 5 years to earn \$20, it does not mean that the investor *gets* exactly \$20; it rather means that the company needs this much time to *earn* as much money as it is invested, and the money earned does not necessary go back to the investors.

Share buyback and P/E ratio P/E ratio can be manipulated by share buybacks since shares are bought with cash and such purchases are not considered expenses under generally accepted accounting practices. The result is reduced shares outstanding and shareholder's equity; in other words, higher P/E and higher ROE, assuming the exact same earnings and returns.

PEG Ratio

PEF ratio is obtained by dividing P/E ratio by estimated growth rate of EPS in percents,

$$\text{PEG} = \frac{\text{P/E ratio}}{\text{Estimated EPS growth rate}}$$

If the EPS of a company is posed to grow 20% and its current P/E ratio is 30, its PEG ratio would be $30 \div 20 = 1.5$.

As is the case of P/E ratio, the lower the PEG ratio is, the better it is. Based on the assumption that earnings growth is linearly proportional to current earnings, a PEG ratio of 1 is fair valued, and a PEG ratio below 1 is undervalued.

P/B ratio

Price-to-book ratio is obtained by dividing share price by *book value per share*. The book value of a company is derived from total assets minus intangible assets and liabilities.

$$\text{Book value} = \text{Total assets} - \text{Intangible assets and goodwill} - \text{Total liabilities}$$

Book value is the amount of money the shareholders get when the company is liquidated. It is thus also known as *shareholder's equity*.

Book value is not a number found in financial statements, but its components are readily available in a balance sheet. Intangible assets, due to their lack of physical existence and subjective value, are often absent in a balance sheet. Goodwill, the excess amount paid during a business acquisition over the fair value of acquired assets, is also an intangible asset but it is listed as a separate item from other intangible assets.

A P/B ratio closer or below 1 is fair or undervalued as a rule of thumb. P/B ratio could be underselling companies that are heavy in intangible assets, such as service provider or software development companies. Brands and patents all are intangible assets. Thus, P/B ratio is a reliable indicator only for the following industries:

- Capital intensive industries
 - Energy
 - Utility
 - Transportation
 - Large manufacturers
 - Mining and refining
 - Semiconductor manufacturing
- Financial services and banks

Banks are usually valued with P/B ratio instead of P/E ratio due to two main reasons. The first reason is also why P/B is favored for utility sector where companies often have a lot of assets as collaterals for loans.

1. Their earnings fluctuate often more than other industries due to federal interest and credit risks.
2. Banks often hold a large amount of loans and liabilities such as deposits and P/B ratio serve as a proxy of a bank's effectiveness on earning.

For example, Warren Buffett mentioned in one of shareholder meetings that he would buy back BRK.B shares when its P/B ratio is below 1.2. However, P/B ratio could still just be a proxy of intrinsic value of

Berkshire Hathaway, as Warren did not buy back any shares in late 2024 when its P/B ratio is still under 1. At the beginning of April 2025 before a correction in the general equity market, P/B ratio of BRK.B reached a historical high of 1.1.

P/FCFE Ratio

Price-to-free cash flow to equity ratio is share price divided by *free cash flow to the equity* (FCFE). Free cash flow to equity, FCFE, or sometimes known as *levered cash flow*, is calculated as such:

$$\text{FCFE} = \text{Operating cash flow} - \text{Capex} + \text{Net debt issued (repaid)}$$

The components of FCFE are readily available in a cash flow statement.

Since FCFE excludes capital expenditure (capex), it is used to fund non-asset related growth, buy back shared, and pay shareholders dividends. A high P/FCFE would mean that an investor is buying a company's cash flow at a premium. As is the case of P/E ratios, P/FCFE ratios are only meaningful when compared with historical P/FCFE values of a company, with the competitors of a company, or with the sector average.

4.2.2 Enterprise Value Multiples

Enterprise Value

Enterprise value (EV) is the total value of a firm, equal to how much it is needed to buy the whole company. It is calculated as such,

$$\begin{aligned} \text{Enterprise Value} &= \text{Market capitalization} \\ &+ \text{Total liabilities} \\ &- \text{Cash and cash equivalents} \end{aligned}$$

where cash and cash equivalents include short-term investments such as T-bills and money market funds but does NOT include marketable securities.

EV serves as a proxy of total capital that a company utilizes to generate profit. It is therefore suitable for other measures of profit at company level or in a wholistic approach, such as operating cash flow and EBITDA.

Negative enterprise value Asian companies tend to hoard a huge pile of cash, effectively rendering their EV negative. Cash at best only earns an interest no higher than bonds, so they are better spent elsewhere. An excess cash could therefore be a red flag that the management team is not good at capital allocation.

EV/EBITDA Ratio

EV/EBITDA ratio is EV divided by *earnings before interest tax depreciation and amortization* (EBITDA). EBITDA is considered a proxy of profitability or cash flow.

EBITDA is not available in income statement, but *earnings before interest and tax* (EBIT, also known as *operating income* OI) often is. EBITDA is simply EBIT plus non-cash expenses, i.e., amortization and

depreciation. If EBIT is not available, we can work from *net income* (NI) or revenue in the following manner:

$$\left. \begin{array}{l} \text{NI} = \text{Revenue} \\ - \text{Cost of Goods Sold} \\ - \text{SG\&A} \\ - \text{R\&D} \\ - \text{Other OpEx} \\ - \text{Interest} \\ - \text{Taxes} \end{array} \right\} \text{Operating Income (EBIT)} = \text{EBITDA} - \text{Depreciation} - \text{Amortization}$$

Use EV/EBITDA the same way as P/E ratio – compare present ratio with historical average, competitors, and sector.

Since EV/EBITDA does not take non-cash expenses and capital structure into consideration, it is suited for the following scenarios and sectors:

- Comparing companies within a same sector but with different debt levels
- Capital intensive sectors (e.g. energy) where debt level and depreciation is huge
- Sectors heavy on intangible assets (e.g. tech) where amortization is huge

A closely related ratio, EV/EBIT, is thus suitable for the following scenarios and sectors:

- Comparing companies with similar debt levels
- Financial firms with heavy debt obligation and tax impact, thus requiring EBIT which gives a clearer picture of capital structure than EBITDA

EV/FCFF Ratio

EV/FCFF is derived EV divided by *free cash flow to the firm* (FCFF). Free cash flow to the firm, FCFF, or sometimes known as *unlevered cash flow*, is calculated as such:

$$\text{FCFF} = \text{NOPAT} + \text{Depreciation and amortization} - \text{Capex} - \Delta\text{NWC}$$

As it turns out, there are more than

Theorem 4.1 FCFF can be obtained from operating cash flow in the following way,

$$\text{FCFF} = \text{Operating cash flow} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Capex}$$

Starting from the definition of operating cash flow, which is earnings with non-cash expenses (depreciation and amortization) added back and change in NWC accounted,

$$\begin{aligned} \text{Operating cash flow} &= \text{Net income} \\ &+ \text{Non-cash expenses} \\ &- \Delta\text{NWC} \end{aligned}$$

Net income is EBIT with tax and interest expensed away,

$$\text{Net income} = \text{Operating income} - \text{Interest expense} - \text{Tax} = (\text{Operating income} - \text{Interest expense}) \times (1 - \text{Effective tax rate})$$

and NOPAT is EBIT with only effective tax rate,

$$\text{NOPAT} = \text{Operating income} \times (1 - \text{Effective tax rate})$$

Thus, net income is NOPAT less a “taxed” interest expense.

$$\begin{aligned} \text{Net income} &= \text{NOPAT} - \text{Interest expense} \times (1 - \text{Effective tax rate}) \\ \Rightarrow \text{Operating cash flow} &= \text{NOPAT} \\ &\quad - \text{Interest expense} \times (1 - \text{Effective tax rate}) \\ &\quad + \text{Non-cash expenses} \\ &\quad - \Delta \text{NWC} \\ &= \text{FCFF} - \text{Interest expense} \times (1 - \text{Effective tax rate}) + \text{Capex} \end{aligned}$$

Theorem 4.2 FCFF and FCFE can be associated in the following way,

$$\text{FCFF} = \text{FCFE} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Net debt issued (repaid)}$$

Recall that FCFE is defined in terms of operating cash flow, and we just demonstrated that FCFF can also be defined in terms of operating cash flow.

$$\begin{aligned} \text{FCFE} &= \text{Operating cash flow} - \text{Capex} + \text{Net debt issued (repaid)} \\ \text{FCFF} &= \text{Operating cash flow} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Capex} \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Operating cash flow} &= \text{FCFE} + \text{Capex} - \text{Net debt issued (repaid)} \\ &= \text{FCFF} + \text{Capex} - \text{Interest expense} \times (1 - \text{Effective tax rate}) \\ \Rightarrow \text{FCFF} &= \text{FCFE} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Net debt issued (repaid)} \end{aligned}$$

That is, FCFF is FCFE, plus the after-tax interests that goes to the debt holders, minus cash from net debts the company raises.

4.2.3 Solvency Ratios

4.2.4 Yields

4.2.5 Return on Invested Capital

Return on Invested Capital

Return on invested capital (ROIC) is a measure of capital allocation and efficiency and defined as:

$$\text{ROIC} = \frac{\text{Net operating profit after tax}}{\text{Invested capital}}$$

Net operating profit after tax (NOPAT) is easy to calculate. It is operating income (OI) taxed on the same basis as pretax income (OI less interest), that is, the theoretical earnings where there is no debt to fund the company (and thus no interest expenses).

$$\text{NOPAT} = \text{Operating income} \times (1 - \text{Effective tax rate})$$

where effective tax rate is derived from income tax and pretax income:

$$\text{Effective tax rate} = \frac{\text{Income tax expense}}{\text{Pretax income}}$$

There is no standard definition for "invested capital." Analysts often make a myriad of adjustments on invested capital to adjust for companies across various scenarios in various industries. The most general formula of invested capital is:

$$\begin{aligned} \text{Invested capital} &= \text{Working capital} \\ &\quad + \text{PP\&E net depreciation} \\ &\quad (+ \text{Intangibles}) \\ &\quad (+ \text{Cash and its equivalents}) \end{aligned}$$

Common adjustments of invested capital include:

- Inclusion/exclusion of intangibles; in particular, goodwill
- Capitalization of intangible investment
- Inclusion/exclusion of cash and current working capital
- Time difference between NOPAT and invested capital

Working capital and excess cash There are two kinds of working capital that measure short-term financial health and operational liquidity of companies.

1. *Net* working capital
2. *Trade* or *non-cash* working capital

Trade working capital focuses more on the trade activities and thus excludes cash items such as prepaid expenses and cash portion of debt.

$$\begin{aligned} \text{NWC} &= \text{Current assets} - \text{Current liabilities} \\ &= \left. \begin{aligned} &\text{Accounts receivable} \\ &+ \text{Inventory} \end{aligned} \right\} \text{Asset portion of TWC} \\ &\quad + \text{Cash and its equivalents} \\ &\quad + \text{Short-term investments} \\ &\quad - \left. \begin{aligned} &\text{Accounts payable} \\ &+ \text{Accrued expenses} \end{aligned} \right\} \text{Liability portion of TWC} \\ &\quad + \text{Current portion of debt and leases} \end{aligned}$$

If we are using NWC in our invested capital, we are met with another issue - excess cash. Cash often is not necessary to operation and just sits there earning a free interest rate, thus it serves more financing role than operating role. Companies keep a high level of cash due to several reasons:

- Young companies need a cushion for their operating activities
- Asian companies tend to hoard a huge pile as their part of culture

As a cashier keeps some cash in his register, *some* cash is necessary for operation. Analysts use 2% to 5% of revenue depending on the life stage of a company. We will use a conservative figure of 5%. Any cash more than that is stripped away in our calculation.

Intangible investment Prior to 2022, intangible assets are only recognized after merge and acquisition (M&A). Any internally generated intangibles are not recognized. Thus, companies such as Microsoft that are heavy on intangibles (brand, marketing, softwares, customer list, etc) appear to have less invested capital, which distorts ROIC.

When making adjustments to the calculation of ROIC, it is import to make sure that the numerator matches the denominator. In this particular adjustment, we are going to capitalize intangible investments and correspondingly adjust amortization item in NOPAT. Start with adjusted NOPAT:

$$\begin{aligned}\text{Adjusted NOPAT} &= \text{EBITA} \\ &\quad + \text{Intangible investment} \\ &\quad - \text{Amortization of all intangibles}\end{aligned}$$

The reason why we start from EBITA instead of EBIT is because part of intangible investment goes into maintaining and replenishing acquired intangibles, the same way capital expenditure covers purchases new PP&E and maintenance of existing PP&E. Leaving amortization in EBIT and subtracting even more amortization from it runs the risk of double counting the amortization of acquired intangibles.

Intangible investment is not a GAAP item, and its subitems, research and development (R&D) and selling general and administration (SG&A), are often embedded in a big line item called operating expense. It is therefore difficult to estimate how much of operating expense goes into intangible investment, especially for companies that do not list R&D and SG&A. For those that do, we can roughly amortize them in the following way:

Item	% going to intangible investment	Useful live in years
R&D	100	6
Sales and marketing	70	2
General administration	20	2

Intangible investment is then added back to EBITA to reflect the fact that a portion of operating expense should be capitalized instead of expensed away. Finally we subtract newly calculated amortization from adjusted EBITA.

For invested capital, it will now have an item that is analog of net PP&E – net intangibles, accumulation of past intangible investments and amortization.

$$\begin{aligned}\text{Net intangibles} &= \text{Intangibles from last year} \\ &\quad + \text{Intangible investment} \\ &\quad - \text{Accumulated amortization}\end{aligned}$$

Needless to say this adjustment is laborious. To get a good estimate of total intangibles of a company we need to go back at least 6 years. The matter is further complicated by laws such as Tax Cuts and Jobs Act (passed in 2017) and Tax Relief for American Families and Workers Act of 2024 which meddles in R&D capitalization.

Luckily the residual value of intangibles are (generally) assumed to be 0, so the total intangibles at year

T is actually the weighted average of intangible investments in the past years.

$$\begin{aligned}\text{Net R\&D intangibles at year } T &= \text{R\&D}_T \\ &+ \text{R\&D}_{T-1} \times \frac{5}{6} \\ &+ \text{R\&D}_{T-2} \times \frac{4}{6} \\ &+ \dots \\ &+ \text{R\&D}_{T-5} \times \frac{1}{6}\end{aligned}$$

In an extreme case where R&D expenditure remains the same for the past 6 years, net R&D intangibles amounts to 3.5 times of R&D expenditure and R&D amortization is equal to R&D expenditure.

Goodwill Goodwill occurs in an M&A event when a company pays more than the fair value for an acquisition. Goodwill is added to the balance sheet and is never amortized away, but it is still subject to impairment tests every year. Say, a company has its fixed assets and intangibles valued at \$80 and is bought at \$100, the goodwill would be \$20. Write-downs will reduce goodwill item in balance sheet.

There are three ways to deal with goodwill:

1. Consider goodwill and goodwill impairment in vested capital
2. Include goodwill and *add back* goodwill impairment under the assumption that the value of goodwill does not wear out and management team must be held accountable for their acquisition decisions
3. Exclude goodwill from invested capital completely

For companies that did a one-time acquisition, it is best to leave goodwill out. For companies whose business model is to acquire other companies frequently, goodwill should be added to invested capital to see if management team overpaying for acquisition is justified. The latter companies usually have a large portion of their assets in goodwill and excluding goodwill will inflate their ROIC immensely.

Deferred tax assets and deferred tax liabilities Deferred tax assets (DTA) are taxes overpaid. They arise from the difference between tax depreciation and accounting appreciation. Deferred tax liabilities (DTL) are tax underpaid. They by no means contribute to operation and should be excluded altogether, although DTA and DTL is usually a small portion of the whole balance sheet that does not have huge impact compared to excess cash and goodwill.

Invested capital calculation We now return to calculate invested capital. We can make a myriad of adjustments here and there, but the most important part is that we are consistent with our numerators and denominators and we are consistent in the way we treat intangibles and working capital across companies across industries. Our calculation is outlined below:

- Intangible investment is *not* adjusted, but we should still pay close attention to subitems of intangibles in financial filling of each company to see if R&D is correctly reflected and amortized.
- Goodwill is included and unimpaired.
- Net working capital is used instead of non-cash working capital
- Cash exceeding 5% of revenue is removed
- DTA and DTL are removed

This leads to the formula of invested capital:

$$\begin{array}{rcl}
 \text{Invested capital} = & \text{Current assets} & \\
 & + \text{Net PP\&E} & \\
 & + \text{Goodwill} & \\
 & + \text{Intangibles} & \\
 & + \text{Other long-term assets} & \left. \vphantom{\begin{array}{l} \text{Current assets} \\ \text{Net PP\&E} \\ \text{Goodwill} \\ \text{Intangibles} \\ \text{Other long-term assets} \end{array}} \right\} \text{Total assets} - \text{DTA} \\
 & - \text{Current liabilities} & \\
 & + \text{DTL} & \\
 & - \text{Excess cash} &
 \end{array}$$

which means that we only need total assets, current liabilities, (DTA and DTL if we are feeling fancy), and revenue to calculate invested capital.

Time difference Assets require time to generate revenue. Newly purchased assets do not return anything right away. The most common time difference used between NOPAT and invested capital is 1 year, meaning that, to calculate ROIC at the end of year 2021, we will use NOPAT at the end of 2021 and assets at the end of 2020. Some sources use *average* invested capital, which is the average of invested capital at the beginning of the period and invested capital at the end of the period.

Example We will use NVIDIA as an example. Financial statements of NVIDIA are listed in Table 11 and Table 12. First we will calculate NOAPT. Using income tax of \$11,146 million and pretax income of \$84,026 million, we get effective tax rate of 13.26%. Using operating income of \$81,453 million, NOPAT is thus \$70,648.

NVIDIA only include acquired intangibles and patents in its intangible item in balance sheet, so it is eligible for intangible investment adjustment. Using R&D and SG&A expenditures from Table 12, we have \$25,477 million of net intangibles from R&D and SG&A ending 2024. The amortization expense in 2025 is 1/6 of R&D expense from 2019 to 2024 and 1/2 of SG&A in 2024, which amount to \$6,395 million. Additionally, amortization expense associated with acquired intangibles and patents is \$593 million as listed on page 65 of 10-K form issued in 2025.

Now look at the balance sheet. NVIDIA's TWC in 2024 is \$12,582 million and its NWC is \$33,714 million. Net PP&E and other assets, not including intangibles, sums up to \$9,760 million. Revenue in 2024 is \$60,922 million, which means that \$4,233 million of cash exceeds 5% of revenue. The exact number of invested capital depends on which adjustment we want to make – to capitalize R&D and SG&A, to use TWC or NWC, to exclude excess cash, or to exclude goodwill.

Table 13 shows ROIC based on various adjustments discussed above. For reference, gurufocus.com gives 174.69% while financecharts.com gives 132.62%, both of which falls in our range of 316% to 108%.

ROIC and WACC ROIC itself tells how effective the management team is at turning assets into profit, but it does not tell if the company increase in value or not as a result of operation. Note that the numerator of ROIC is NOPAT, which is to be returned to debt issuers and shareholders, we compare it with *weighted average cost of capital* (WACC). The exact value of WACC depends on sensitivity of stock price, risk free rate, debt-to-equity ratio, and many other factors, but 10% is usually a good conservative estimate, based on 9.5% CAGR of S&P500. If ROIC is greater than 10%, it is likely that the company is creating value.

ROIC, reinvestment rate ROIC itself measures capital efficiency, but it is not necessary the rate at which the company grows. As is the case of Asian companies, even if they have very efficient capital allocation and utilization, they won't grow if they don't reinvest their huge pile of cash. As such, the growth of earnings is co-determined by ROIC and *reinvestment rate*.

Revenue	130,497
Gross profit	97,858
Operating expense	
Research and development	12,914
Sell, general and administration	3,491
Total operating expense	16,405
Operating income	81,453
Interest income	1,786
Interest expense	(247)
Other, net	1,034
Other income (expense), net	2,753
Income before income tax	84,026
Income tax expense (benefit)	11,146
Net income	72,880

(a) Excerpt of NVIDIA's income statement of FY2025, in millions.

Current assets:	
Cash and cash equivalents	7,280
Marketable securities	18,704
Accounts receivable, net	9,999
Inventories	5,282
Prepaid expenses and other current assets	3,080
Total current assets	44,345
Properties and equipment, net	3,914
Operating lease assets	1,346
Goodwill	4,430
Intangible assets, net	1,112
Deferred income tax assets	6,801
Other assets	4,500
Total assets	65,728

Current liabilities:	
Accounts payable	2,699
Accrued and other current liabilities	6,682
Short-term debt	1,250
Total current liabilities	10,631
Total liabilities	22,750

(b) Excerpt of NVIDIA's balance sheet of FY2024, in millions.

Table 11: Financial statements of NVIDIA.

Reinvestment Rate

Reinvestment rate (RR) is the ratio of increase in net PP&E and NWC (and net intangibles) to NOPAT. It is essentially the portion of earnings that goes into capital expenditures and investments.

$$\text{Reinvestment rate} = \frac{\text{Capex} + \Delta\text{NWC} - \text{Depreciation}}{\text{NOPAT}}$$

As we will see in the following text, the value of a company (be it enterprise value or equity value) is (theorized to be) the sum of present values of future cash flows. It is therefore crucial to project cash flow growth, and ROIC is just a few steps away from growth projection.

$$\text{Expected EBIT growth} = \text{ROIC} \times \text{RR}$$

Assuming constant RR (i.e. the portion of capex to OCF does not change), constant portion of expenses, and constant portion that goes into change of NWC, cash flow growth is the same as earnings growth. This FCF growth estimate can be either used to project future cash flow in a DCF model, or to compare with implied future growth in a reverse DCF model.

When depreciation and decrease in NWC is greater than capital expenditure, reinvestment rate goes negative. Two main reasons why a company has negative reinvestment are,

1. The company over-invested in the past and is living off past investment.
2. The company is liquidating itself and letting assets wear down.

For the first case, we should use industry average. For the second case, however, there is no growth to

FY	R&D	SG&A
2025	12,914	3,491
2024	8,675	2,654
2023	7,339	2,440
2022	5,268	2,166
2021	3,924	1,940
2020	2,829	1,093
2019	2,376	991
2018	1,791	815

Table 12: R&D and SG&A expenditures of NVIDIA since fiscal year 2018.

Without in-house intangibles	TWC	NWC, excluding excess cash	NWC, including excess cash
Excluding goodwill	316.23%	180.05%	162.52%
Including goodwill	263.90%	161.78%	147.49%
With in-house intangibles	TWC	NWC, excluding excess cash	NWC, including excess cash
Excluding goodwill	166.98%	123.38%	115.81%
Including goodwill	152.82%	115.48%	108.81%

Table 13: ROIC of NVIDIA in fiscal year 2025.

project. The company is digging its grave.

Take, again, NVIDIA for example. Given its capex, investments, depreciation and amortization, and change in NWC, we conclude that its reinvestment rate is 50.48%. Combined with the ROIC of 115.48%, its EBIT is expected to grow at 58.3%.

Capex	3,236
Intangible investment	16,405
Depreciation	1,300
Amortization of in-house intangibles	6,395
Δ NWC	28,365
NOPAT adjusted for in-house intangibles	79,849
Reinvestment rate	50.48%
ROIC,	
adjusted for in-house intangibles,	
using NWC less excess cash,	115.48%
including good will	
Expected EBIT growth	58.3%

Table 14: Reinvestment rate and expected EBIT growth of NVIDIA at the end of fiscal year 2025. Numbers are in millions except for percentages.

4.2.6 Return on Equity

Return on Equity

Return on equity (ROE) measures how much earnings a company generates from shareholders' equity.

$$\text{ROE} = \frac{\text{Net income}}{\text{Shareholder's equity}}$$

Equity Reinvestment Rate

Equity reinvestment rate (ERR) measures how much of earnings is reinvested back to capital.

$$\text{ERR} = \frac{\text{Capex} + \Delta\text{NWC} - \text{Depreciation} - \text{Net debt issued (repaid)}}{\text{Net income}}$$

ROE and ERR are analogues of ROIC and RR on the equity side. Likewise, cost of equity (CoE) is used in the place of WACC. Different from NOPAT and ROIC which ignores capital structure, on the equity side, we must take capital structure into consideration as well as many other equity side shenanigans:

1. Earnings can be raised by issuing shares.
2. Earnings can be raised by issuing debt.
3. Leverage (total assets to shareholders' equity) inflates ROE.

While the effect of share dilution can be eliminated by calculating EPS, the effect of debt financing must be accounted by adjusting net debt issued (repaid) in equity reinvestment rate.

4.2.7 Determinants of Ratios

4.3 Fair Value Calculation

Based on the EMH that the price accurately reflects all the cash flow in the future, we can project cash flow of a company, discount future cash flow by some discount rate, and arrive at a "fair" price for its shares. The calculation of fair value is thus similar to that of bonds; in the case of valuing a company, we replace par with *terminal value* and coupon with *free cash flow* in the equation. This method of valuing a company is thus called *discounted free cash flow model* (DCF).

Other valuation models exist such as:

- Reverse discounted free cash flow model
- Dividend discount model
- Excess returns model

Since DFC is not suitable for all companies, these models are necessary valuation tools. For example, dividend discount model are suitable for companies that pay out a great portion of their cash flow as dividends; excess returns model suit financial companies such as banks and insurance [16].

4.3.1 Discounted Cash Flow Model

Components of DCF The calculation of DFC is based on the following four components:

- Time horizon N
- Free cash flow projection, with the following varieties of cash flow available:
 - Free cash flow *to the firm*, FCFF, or *unlevered* free cash flow
 - Free cash flow *to equity*, FCFE, or *levered* free cash flow
- Discount rate r , with the following choices available:
 - Weighted average cost of capital (WACC), which is appropriate for FCFF
 - Cost of equity (CoE) or required rate of return (RRR), which is appropriate for FCFE
 - Custom discount rate such as 10-year treasury bond yield, GDP growth rate, or CAGR of S&P 500, sometimes with a margin of safety applied
- Terminal value (TV), with the following models available:
 - Gordon growth model, which uses a perpetuity growth rate g
 - Exit multiple approach, which assumes a valuation multiple at the end of time horizon

The present value (PV) of future cash flow FCF_i at year i is discounted by the power of $1 + r$.

$$PV_i = \frac{FCF_i}{(1 + r)^i}$$

The sum of PVs of future FCF and PV of TV is *enterprise value* (EnV). The sum of PVs of future FCFE and PV of TV is *equity value* (EqV). EnV and EqV are related in the following way:

$$EnV = \sum_{t=1}^N \frac{FCFF_t}{(1 + r)^t} + \frac{TV}{(1 + r)^N} \quad (8)$$

$$EqV = \sum_{t=1}^N \frac{FCFE_t}{(1 + r')^t} + \frac{TV}{(1 + r')^N} \quad (9)$$

$$= EnV - \text{Total liabilities} + \text{Cash and its equivalents and marketable securities} \quad (10)$$

Free cash flow projection Free cash flow projection contains a lot of bias and guesswork that a consensus between various analysts must be used; if there is no estimation of future free cash flow from analysts available, an approach is to interpolate free cash flow growth rate starting from the mean of last reported growth rate and historical average growth rate and to reduce it gradually and at the end of time horizon it reaches stable growth rate g , which is often assumed to be the yield of 10-year government bond, or an average of the yield of 10-year government bond for the last 5 years [16].

Terminal value In Gordon growth model, the same stable growth rate g is also used to calculate TV. Given the stable growth rate g , the appropriate discounting rate r for FCF, the model gives an approximate value of TV,

$$TV = \frac{FCF_N \cdot (1 + g)}{r - g} \quad (11)$$

TV can be otherwise estimated by multiplying the earnings at the end of time horizon with an exit multiple. For FCFF, TV would be the product of estimated EBITDA and estimated EV/EBITDA. For FCFE, TV would be the product of estimated net income and estimated P/E ratio. Exit multiples are often derived from multiples of comparable companies and multiples of precedent transactions of comparable targets.

Weighted average cost of capital and cost of equity WACC measures the expected payment on average to all its security holders (stockholders and debt issuers) to finance its assets. Since companies are financed on multiple sources such as stocks, debts, subsidies, WACC measures the minimum rate of return to satisfies their investors. Suppose that a company is financed solely by debts and stocks,

$$WACC = \frac{E}{D + E} \cdot R_E + \frac{D}{D + E} \cdot R_D \cdot (1 - T) \quad (12)$$

where D is total liability, E is total shareholder equity, R_D is *cost of debt*, R_E is *cost of equity*, and T is tax rate which is imposed on debt interest and is usually 30%. Both D and E can be found in a company's balance sheet.

Cost of equity R_E can be calculated using CAPM:

$$R_E = r_f + \beta \cdot (r_m - r_f)$$

where r_f is risk-free rate, β is *levered beta*, and r_m is expected equity return. Again, we are using average

10-year government bond yield for r_f . The term $r_m - r_f$ is known as *excess equity return* or *equity risk premium*, which can be sourced from Aswath Damodaran's website [7]. The implied equity risk premium is 4.12% on March 1st, 2025, for example.

An advanced version of CoE calculation is based on Fama-French 3-factor model:

$$R_E = r_f + \beta \cdot (r_m - r_f) + b_s \cdot \text{SMB} + b_v \cdot \text{HML} + \alpha$$

where SMB is size premium, historical excess return of small caps over big caps, and HML is value premium, historical excess return of high book-to-market ratio companies to low ratio companies. Alpha is an error term. Both b_s and b_v are obtained by linear regression.

Cost of debt R_D is calculated from annual interest expense and total liability [17]. Annual interest expense can be found in the income statement.

$$R_D = \frac{\text{Annual interest expense}}{D}$$

The beta term β measures the company's sensitivity to market volatility and takes its capital structure (debt-equity ratio) into consideration. Let β_0 be the unlevered beta for an industry, the levered β is defined as

$$\beta = \beta_0 \cdot \left[1 + (1 - T) \cdot \frac{D}{E} \right]$$

where again T is tax rate. For companies outside Canada and The United States, we use the North American industry average unlevered beta [16]. Unlevered betas can be found on Aswath Damodaran's website [6].

Equity value The last step of DCF is to subtract net debt (debt D minus cash and cash equivalents C minus marketable securities M) from enterprise value. The result is called *equity value* (EqV) which represents the calculated fair value of the company's equities. Divide this number by shares outstanding and we arrive at the "fair" price of a share.

$$\text{Fair share price} = \frac{\text{EqV}}{\text{Shares outstanding}} = \frac{\text{EnV} + M + C - D}{\text{Shares outstanding}}$$

Example Take NVIDIA for example. We will be using data from 2020 and calculate its cash flow projection as well as fair value based on our assumptions as list in Table 15. For beta we will use the value from 2021 since beta from previous periods are unavailable as of 2025.

We start from levered beta β :

$$\beta = 0.96 \times (1 + 0.7 \times 11898 \div 12204) = 1.615$$

Cost of equity R_E is thus:

$$R_E = 0.02 + 1.615 \times 0.0472 = 0.0962$$

To calculate projected free cash flow, we must first obtain FCFE growth rate g_i . The growth rate of FCFE at first year g_1 will be the mean of historical average 60% and last growth rate 36%, which amounts to 48%. The growth rates on the last year of time horizon g_{10} equals to risk free rate of 2%. Growth rates for the years in the middle will be interpolated using these two numbers. Growth rates and FCFE and discounted FCFE for each year is listed in Table 16. The sum of all discounted FCFE is thus \$112,676 million.

Item	Value
Historical average growth rate of free cash flow to equity	60%
Free cash flow (2020/01/31)	\$4,272
Free cash flow annual growth rate (2020/01/31)	36%
Interest expense (2020/01/31)	\$52
Total liability D (2020/01/31)	\$11,898
Shareholder equity E (2020/01/31)	\$12,204
Cash, cash equivalents, and marketable securities C	\$10,897
Shares outstanding (2020/01/31)	618
Unlevered beta for semiconductor industry β_0	0.96
Equity risk premium (2020)	4.72%
5-year average of 10-year U.S. treasury (2016 Jan. – 2020 Jan.) $r_f = g$	2%
Time horizon N	10 years

Table 15: Data from NVIDIA 2020 financial statement as well as other sources [2] [7] [10]. All numbers are in millions except for percentages and time horizon.

Year i	Growth rate g_i (%)	FCFE $_i$ (million \$)	Present value of FCFE $_i$ (million \$)
1	48	6,322	5,767
2	38.8	8,775	7,303
3	34.2	11,777	8,940
4	29.6	15,263	10,570
5	25	19,078	12,053
6	20.4	22,970	13,238
7	15.8	26,600	13,984
8	11.2	29,579	14,186
9	6.6	31,531	13,795
10	2	32,162	12,836

Table 16: FCFE and FCFE growth rate according to DFC assumptions.

Terminal value will be based on the last FCFE which is \$12,836 million.

$$TV = \frac{12,836 \times 1.02}{0.0962 - 0.02} = 171,820$$

$$\text{Present value of TV} = 171,820 \div 1.0962^{10} = 68,576$$

Finally, we arrive at enterprise value and equity value. Since our calculated enterprise value is way larger than cash and debt at 2020, we ignore them and directly derive “fair” share price from enterprise value.

$$EqV \approx EnV = 112,676 + 68,576 = 181,252$$

$$\text{Fair share price} = 181,252 \div 618 = 293.29$$

On Jan. 20th 2020, NVIDIA trades at \$250.4 per share (this price level was before a 4:1 split in 2021 and a 10:1 split in 2024). Calculation shows that this price level is below fair value; that is, NVIDIA is undervalued.

Limitations of DCF It goes without saying that DFC as a prediction model is as good as its assumptions and its inputs. The projection of cash flow could be wrong, the CAPM we used in the calculation of discount rate is also empirically wrong, and the market can simply disregard a calculated “fair” value. Common practices include

- Slap a 20% margin of error to the calculated fair value; that is, if the current share price is lower than 20% of calculated fair price, we can confidently say that the company is undervalued.
- Prepare different FCFE projections; the best case, the worst case, and the normal case..

4.3.2 Reverse Discounted Cash Flow Model

Since FCF projection remains the most uncertain and the most influential part of FCF, we can instead start from the present-day share price and reverse engineer the required cash flow to sustain that price level. If the result required future cash flow is unrealistically high, it implies that the current share price is too high and it prices more about hype than future cash flow.

Modeling Each Year's FCFE Reverse-engineering future cash flow implies that there is only one variable among future cash flow. The most common assumption to make is that the FCFE growth remains constant over the years ($g_1 = g_2 = \dots = g_{N-1}$). We can also assume that the growth rate starts from an unknown value g_1 and declines at the same rate over the years until it reaches terminal growth rate g , the same assumption we did previously for DCF. Thus,

$$\begin{aligned} \text{For } i = 2 \dots N, g_i &= g_1 \cdot \frac{N - i + 1}{N} + g \cdot \frac{i - 1}{N} \\ \text{For } i = 1 \dots N, \text{FCFE}_i &= \text{FCFE}_0 \prod_{j=1}^i (1 + g_j) \\ \text{TV} &= \frac{\text{FCFE}_N \cdot (1 + g)}{R_E - g} = \frac{1 + g}{R_E - g} \cdot \text{FCFE}_0 \prod_{j=1}^N (1 + g_j) \end{aligned}$$

where FCFE_0 is the last reported FCFE.

With all variables depending solely on g_1 , we can use trial error to approximate g_1 that matches a share price of \$250.4, a terminal growth rate of 2%, and a CoE of 9.62%, the same set of data and assumptions based on Table 15. The value of g_1 turns out to be 32.9%, which is below last reported FCFE growth rate. If one expects FCFE of NVIDIA to grow at the same pace as last reported, the FCFE growth of 32.9% implied by current share price would be way below his expectation, and NVIDIA could be undervalued.

4.3.3 ROIC and Discounted Cash Flow

Recall that, when we calculate terminal value, Gordon growth model calls for a growth rate. Previously we also learned that growth is tied to ROIC and reinvestment rate. It is therefore crucial to incorporate ROIC into our DCF model as it gives us more insight. Let us preface our analysis of incorporation of ROIC with the following assumptions and generalizations:

- Our DCF model consists of only terminal stage here. The company is in a mature stage and grows at a perpetual growth rate of $g = \text{ROIC} \times \text{RR}$.
- Interest ratio I and effective tax rate T remains the same, so that EBIT and earnings E grow at the same rate. In particular, $E = \text{EBIT} \times (1 - I) \times (1 - T)$.
- Cash conversion rate (CCR) remains the same, so that earnings and FCFE grow at the same rate. In particular, $\text{FCFE} = E \times \text{CCR}$.

These assumptions allow us to apply growth (and eventually ROIC) to earnings and convert it to free cash flow conveniently. Let E' be the earnings at next year and FCFE' be the FCFE at next year. Since FCFE

is what is left after reinvestments are made, we have,

$$\begin{aligned}\text{FCFE}' &= E' \times \text{CCR} \times (1 - \text{RR}) \\ &= E' \times \text{CCR} \times \left(1 - \frac{g}{\text{ROIC}}\right) \\ &= E \times (1 + g) \times \text{CCR} \times \left(1 - \frac{g}{\text{ROIC}}\right)\end{aligned}$$

Gordon growth model determines the value of a company based on the free cash flow next year. Therefore, given the required rate of return r ,

$$\text{Value} = \frac{\text{FCFE}'}{r - g} = \frac{E \times \text{CCR} \times (1 + g) \left(1 - \frac{g}{\text{ROIC}}\right)}{r - g}$$

Dividing the value by earnings E gives a “fair” P/E ratio,

$$\text{Fair P/E ratio} = \frac{\text{Value}}{E} = \frac{\text{CCR} \times (1 + g) \left(1 - \frac{g}{\text{ROIC}}\right)}{r - g}$$

Based on different values of growth g and ROIC spread (ROIC minus WACC), we get a fair P/E ratio matrix as shown in Table 17. Notice that when ROIC spread is 0%, fair P/E ratio barely moves as growth varies. When ROIC spread is negative, the more growth the company pursues, the less P/E ratio it becomes even to the point where it could go negative. This is because, with an inefficient capital utilization, the company must reinvest more than it can afford to achieve a high growth.

Growth \ ROIC spread		ROIC spread						
		−3%	−2%	−1%	0%	1%	2%	3%
	2%	9.11	9.56	9.92	10.20	10.43	10.63	10.79
	3%	8.41	9.20	9.81	10.30	10.70	11.04	11.32
	4%	7.43	8.67	9.63	10.40	11.03	11.56	12.00
	5%	6.00	7.88	9.33	10.50	11.45	12.25	12.92
	6%	3.79	6.63	8.83	10.60	12.05	13.25	14.27
	7%	0.00	4.46	7.93	10.70	12.97	14.86	16.46
	8%	−7.71	0.00	6.00	10.80	14.73	18.00	20.77

Table 17: Fair P/E ratio based on different growth and ROIC spread, assuming $\text{CCR} = 1$.

The table also points out an interesting property of P/E ratio. Under this simple model or a full-fledged DCF model, P/E ratio itself does not imply just growth. A P/E ratio of 10.8 could imply an ROIC spread of 3% and a growth of 2% (high profitability and low growth), or an ROIC spread of 0% and a growth 8% (low profitability and high growth).

The relation between growth and ROIC now proposes two issues with our established DCF model:

1. The growth in stable stage still comes with a cost that we cannot ignore.
2. The free cash flow number the terminal value based on does not reflect change in reinvestment rate and growth

For the discussion below we assume that ROIC stays constant in any stages. We also assume, without loss of generality, interest expense and taxes are 0 such that EBIT equals to NOPAT and it can be converted to FCFF directly through $1 - \text{RR}$.

Use FCF at year $N + 1$ In our established DCF model, our TV is based on FCF_N where N is length of growth stage. In the most simple two-stage DCF, FCF grows at a growth rate of g' at growth stage and $g = 2\%$ at stable stage. Recall that TV is defined in Equation 11, which is actually an infinite geometric series simplified into one term,

$$TV = FCF_N + FCF_N \cdot \frac{1+g}{1+r} + FCF_N \cdot \left(\frac{1+g}{1+r}\right)^2 + \dots$$

The problem arises when we use ROIC and RR to determine growth rate of EBIT instead of FCFF. With the introduction of RR, we also have to assume different RRs at growth and stable stage, which means that EBIT at year $N + 1$ will grow at a different rate than FCFF, because FCFF is ultimately determined by reinvestment, not FCFF from the previous year.

$$\begin{aligned} \frac{EBIT_N}{EBIT_{N-1}} &= 1 + g' = 1 + ROIC \cdot RR' \\ \Rightarrow \frac{FCFF_N}{FCFF_{N-1}} &= \frac{EBIT_N \cdot (1 - RR')}{EBIT_{N-1} \cdot (1 - RR')} = 1 + g' \\ \frac{EBIT_{N+1}}{EBIT_N} &= 1 + g = 1 + ROIC \cdot RR \\ \Rightarrow \frac{FCFF_{N+1}}{FCFF_N} &= \frac{EBIT_{N+1} \cdot (1 - RR)}{EBIT_N \cdot (1 - RR')} = (1 + g) \cdot \frac{1 - RR}{1 - RR'} \end{aligned}$$

Therefore, TV must be based on FCF at year $N + 1$ instead of N , and FCF at year $N + 1$ must be recalculated according to stable RR and growth instead of extrapolated using RR' and g' .

By using FCF at year $N + 1$, we also have the term $1 - RR$ baked into our TV. With $RR = g/ROIC$, the requirement that ROIC is greater than WACC is still reflected in TV.

5 RSU and ESPP

5.1 RSU

Restricted Stock Unit

Restricted stock units (RSUs) are stock shares employees can earn full ownership after a period of time.

Vest

Stock *vests* when some predefined conditions are met and an predefined amount of stock shares are granted to an employee.

Vesting schedule

Vesting schedule refers to how RSUs are granted to an employee over the time the employee stays at the company.

There are 3 types of vesting schedule: